**Name** _____ Period _____ Date _____

# 3-5 Lesson Quiz: Zeros of Polynomial Functions

short form

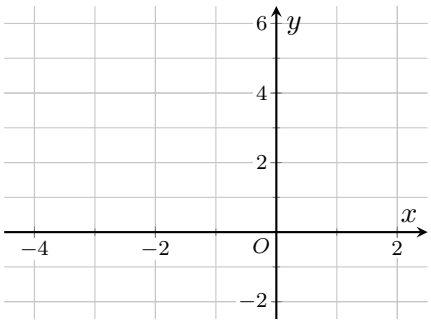
**1. Part A** What are all the zeros of $f(x) = x^3+2x^2-3x$? Show your factoring:

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_____

Zeros: _____

**Part B** Use the zeros to sketch the graph of $f$.




**4.** What are the zeros and their multiplicities for the function $y = x^3 + 3x^2 + x + 3$, shown in the graph?

Ⓐ $-3$, $-1$, and 1, each multiplicity 1

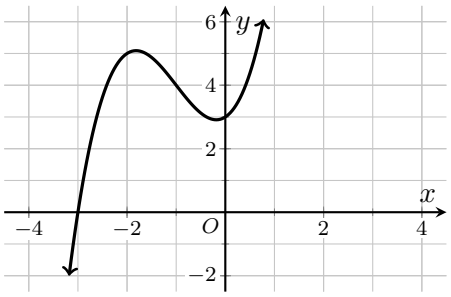
Ⓑ $-3$, multiplicity 1

Ⓒ 3, multiplicity 2

Ⓓ $-3$, multiplicity 3

How does the graph (or the equation) tell you?

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**5.** Find the zeros of $f(x) = -x^3 - 2x^2 + 7x - 4$. Then describe the behavior of the graph of $f$ at each zero.

Ⓐ 4, $-1$; As $x \to -\infty$, $f \to -\infty$. When $-1 < x < 4$, $f < 0$. At $x = 4$, $f$ is tangent to the $x$-axis, so when $x > 1$, $f \to \infty$.

Ⓑ $-4$, 1; As $x \to -\infty$, $f \to -\infty$. When $-4 < x < 1$, $f > 0$. At $x = 1$, $f$ is tangent to the $x$-axis, so when $x > 1$, $f \to -\infty$.

Ⓒ 4, $-1$; As $x \to -\infty$, $f \to \infty$. When $-1 < x < 4$, $f > 0$. At $x = 4$, $f$ is tangent to the $x$-axis, so when $x > 1$, $f \to -\infty$.

Ⓓ $-4$, 1; As $x \to -\infty$, $f \to \infty$. When $-4 < x < 1$, $f < 0$. At $x = 1$, $f$ is tangent to the $x$-axis, so when $x > 1$, $f \to -\infty$.

Show your work (test a zero, divide, factor):

_____

_____

**Scoring:** Question 1 (Parts A and B together) and Question 5 are 10 points each. Question 4 is a bonus of up to 5 points. Total out of 20. (Numbers match the Savvas Lesson Quiz; Questions 2 and 3 are not on this form.)

# 3-5 Lesson Quiz, short form: answer key

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1. **Part A: $-3$, 0, 1.** $f(x) = x(x^2 + 2x - 3) = x(x + 3)(x - 1)$. Savvas item 1A, DOK 2, Realize skill F62. **Part B:** a cubic rising to the right that crosses the $x$-axis at $-3$, 0 and 1: below the axis for $x < -3$, above on $-3 < x < 0$ (local maximum near $(-1.9, 6.1)$), below on $0 < x < 1$ (local minimum near $(0.5, -0.9)$), above for $x > 1$. Savvas item 1B, DOK 2, Realize skill A73. Taught by Example 1 and Try It 1a. **10 points:** Part A 4 (2 for the factored form, 2 for all three zeros); Part B 6 (2 for crossing at the three zeros, 2 for the correct side of the axis on each interval, 2 for the end behavior).

4. **B, $-3$, multiplicity 1.** By grouping, $x^3 + 3x^2 + x + 3 = x^2(x + 3) + (x + 3) = (x + 3)(x^2 + 1)$. The factor $x + 3$ appears once, and the graph crosses at $-3$ with no flattening; $x^2 + 1$ gives $\pm i$, which are not $x$-intercepts. Distractor A invents $\pm 1$; D would flatten at $-3$. Savvas item 4, DOK 1, Realize skill F62. Taught by Example 2 (multiplicity) with factoring by grouping. **Bonus 5:** B with a reason (the factored form, or "crosses once without flattening"); 3 for B alone.

5. **D.** $f(1) = -1-2+7-4 = 0$; synthetic division by 1 gives $-x^2-3x+4$, so $f(x) = -(x-1)(x^2+3x-4) = -(x-1)^2(x+4)$. Zeros $-4$ (multiplicity 1, crosses) and 1 (multiplicity 2, tangent). The leading term $-x^3$ gives $f \to \infty$ as $x \to -\infty$ and $f \to -\infty$ as $x \to \infty$; $f(0) = -4 < 0$, so $f < 0$ on $-4 < x < 1$. Savvas item 5, DOK 2, Realize skill F62. Taught by Examples 1 and 2; rehearsed by the packet Practice. **10 points:** 6 for D, 4 for the work (a zero tested and the factored form, or $f(1) = 0$ and $f(-4) = 0$ checked).

**Items left off this form** (available for a retake or Form B): Savvas item 2, solve $x^3+5x^2-x-7 = x^2+6x+3$ (answer $x = -5$, $x = -1$, $x = 2$; DOK 2, F62; Example 5, cut); Savvas item 3, solve $x^3 - 4x > 0$ (answer $-2 < x < 0$ or $x > 2$, the union of the two intervals; the TE prints "and"; DOK 2, F62; Example 6, cut).

**Graphs:** the Question 4 graph is redrawn from the TE description and the function itself (crosses at $-3$, turning points near $(-1.8, 5.1)$ and $(-0.2, 2.9)$, $y$-intercept 3). The Question 1 Part B answer graph is not printed; students sketch on a blank grid.

**IXL replacement:** SmartScore 70 or above on PVM, 8CQ and NCY replaces this quiz with 20/20.